

AI MUSIC PRODUCTION PLATFORM

COMPLETE INSTALLATION COMPLETION PLAN

FINAL INSTALLATION PHASE — NO MUSICAL DEVELOPMENT YET

OBJECTIVE .0

המטרה של המשימה הזאת היא לסיים לחלוטין את כל שכבת ההתקנות וה־provisioning של הפרויקט.

בסוף המשימה לא צריך להיות שום מודל או ספרייה מהרשימה המאושרת שנמצאים במצב עמום כמו:

"almost installed"

"worker exists"

"needs checking"

"configured"

"probably ready"

כל רכיב חייב להגיע לאחד מהמצבים הבאים בלבד:

READY

RESEARCH_READY

BLOCKED_LICENSE

BLOCKED_NO_WEIGHTS

BLOCKED_UPSTREAM

BLOCKED_MISSING_LICENSED_ASSET

אין מצב ביניים אחר בסיום המשימה.

HARD RULES .1

במהלך המשימה הזאת:

אסור לבצע:

Golden Song evaluation -

Arrangement Director V2 -

Phrase Planner -

CompositionEngine V2 -

Candidate Ranking logic -

UI redesign -

cost optimization -

export hardening -

additional logging infrastructure שאינו נדרש להתקנה -

- new provider discovery
- שינוי מודלים שכבר עובדים
- upgrade של runtimes קיימים ללא צורך מוכח

מותר לבצע רק:

```
install
bootstrap
provision
pin
hash
license
build
deploy
health
smoke
attest
promote
connect
catalog
```

המטרה היא:

«.Finish installation first»

2. FIRST ACTION — CREATE INSTALLATION MATRIX

לפני כל עבודה נוספת צור:

installation-matrix-v2.json

זה יהיה Source of Truth יחיד לכל ההתקנות.

:Schema

```
}
  , "" : "provider"
  , "" : "category"
  , "" : "codeRepository"
  , "" : "codeRevision"
  , "" : "modelRepository"
  , "" : "modelRevision"
  , runtimeBuilt": false"
  , sourcePinned": false"
  , assetsDownloaded": false"
  , assetsChecksummed": false"
```

```

,assetManifestCreated": false"
,volumeProvisioned": false"
,"" : "licenseStatus"
,secretsConfigured": false"
,realSmokePassed": false"
,nonSilentOutputVerified": false"
,endpointDeployed": false"
,endpointConfigured": false"
,healthReady": false"
,promotionRequired": false"
,promotionSigned": false"
,apiConnected": false"
,"" : "finalStatus"
,[], : "blockers"
[] : "notes"
{

```

VALID FINAL STATUSES .3

:Use only

```

READY
RESEARCH_READY
BLOCKED_LICENSE
BLOCKED_NO_WEIGHTS
BLOCKED_UPSTREAM
BLOCKED_MISSING_LICENSED_ASSET

```

:Do NOT use

```

configured
mostly ready
pending
partial
experimental

```

. "in "finalStatus

.Those may appear only during execution

ABSOLUTE READY DEFINITION .4

:A provider is "READY" only if

source revision pinned
runtime built
model revision pinned
assets downloaded
assets hashed
assets on durable storage
license recorded
required secrets configured
real inference executed
output non-empty
output non-silent where audio
output not source-copy where generation
provider identity verified
checkpoint identity verified
health endpoint reports truthful READY
endpoint deployed
API configured

:For GPU providers that use promotion

real GPU smoke
+
signed promotion
+
endpoint identity matching promotion

.are mandatory

FREEZE EXISTING VERIFIED STACK .5

:Do NOT rebuild unnecessarily

ACE_STEP
BS_ROFORMER
MT3
ALL_IN_ONE

BASIC_PITCH
DEMUCS

MADMOM
TORCHCREPE
ESSENTIA
CHROMA
PYLOUDNORM

.Only verify that the current working deployment remains intact

SHEETSAGE — FINISH AND CLOSE .6

:Current status

.SheetSage is already one of the most advanced new providers

:It already has

- worker
- runtime
- licensed assets
- endpoint
- API wiring
- melody output
- harmony output
- timing evidence
- provenance
- real inference architecture

:Endpoint currently configured

SHEETSAGE_API_URL

.License authorization exists

Remaining work

Run real live health 6.1

:Verify

- runtimeReady=true
- checkpointReady=true
- smokeTested=true
- status=ready

Verify signed persistent smoke proof 6.2

:The proof must bind

- asset manifest
- runtime fingerprint
- package contents
- fixture

output
model identity

Verify API path 6.3

:Run

real audio
SheetSage →
melody →
chords →
timing →
Node analysis provider →

Final status

:If all pass

SHEETSAGE = RESEARCH_READY

.because weights remain non-commercial/share-alike

.Do not remove licensing metadata

BEAT THIS — FINISH IMMEDIATELY .7

.Current state is already advanced

:Pinned

beat-this==1.1.0
checkpoint final0
source commit pinned
checkpoint SHA256 known
Python 3.11
Torch 2.5.1
L4

:Endpoint already exists

MUSIC_PROVIDER_BEAT_THIS_URL

Required actions

.Run checkpoint bootstrap .1
.Verify SHA256 .2

.Run real music audio .3
:Verify .4
at least two beats -
valid monotonically increasing timing -
valid downbeats -
plausible BPM -
.Persist smoke proof .5
.If promotion policy applies, create signed promotion .6
.Verify Node parser accepts real response .7
.Confirm Beat This is scheduled as primary beat evidence .8

:Final

BEAT_THIS = READY

MR-MT3 — COMPLETE INSTALLATION .8

:Current problem

checkpoint_sha256 = null
checkpoint_upstream_revision = null

.Mutable upstream identity is NOT acceptable

Required

Obtain exact checkpoint 8.1

.Download in isolated provider volume only

:Compute 8.2

bytes
SHA256

Resolve immutable source identity 8.3

: "If upstream only uses mutable "main"

.record exact downloaded artifact SHA and repository commit/snapshot evidence

.Never pretend "main" is immutable

Run real GPU smoke 8.4

.Input must contain actual musical notes

:Required output

terminated MIDI note 1 =<

.Zero-note output = FAIL

Bind compatibility patch 8.5

:Persist

adapter source SHA

patch SHA

checkpoint SHA

runtime version

Signed promotion 8.6

:Create independent promotion for

MR_MT3

Deploy endpoint 8.7

:Example canonical env

MR_MT3_API_URL

:Final

MR_MT3 = READY

.only after promotion and live health

YOUR-MT3 — COMPLETE INSTALLATION .9

.Same discipline as MR-MT3

:Current blockers

checkpoint_sha256=null

mutable upstream source

Required

."Provision exact "last.ckpt .1

- .Calculate SHA256 .2
- .Record full artifact tree identity .3
- .Apply only reviewed compatibility patch .4
- .Run actual GPU transcription .5
- .Require non-zero notes .6
- .Verify 8-stem/multitask output where applicable .7
- .Persist execution provenance .8
- .Sign promotion .9
- .Deploy endpoint .10
- .Connect Node parser .11

:Final

YOUR_MT3 = READY

SONGFORMER — FINISH FULL INSTALL .10

:Current manifest explicitly says

NOT_READY_PENDING_BOOTSTRAP_AND_REAL_AUDIO_SMOKE

Existing pinned source

.Keep current exact Git revision

.Keep current HF revision

:Required assets include

SongFormer.safetensors

SongFormer.pt

MusicFM/pretrained_msd.pt

MusicFM/msd_stats.json

MuQ dependency assets if required

Required actions

.Provision every asset .1

.Convert upstream MD5 evidence into local SHA256 inventory .2

:Persist .3

bytes -

SHA256 -

source -

revision -

.Verify submodules are exact .4

.Build runtime .5

.Run real full-song audio .6
:Validate .7
section boundaries -
section labels -
no empty output -
.Store smoke proof .8
.Deploy Modal endpoint .9
:Add .10

SONGFORMER_API_URL

.Connect to analysis provider layer .11
.Do NOT remove All-In-One yet .12
:Final

SONGFORMER = READY

.if licensing supports intended production use

MOSS-MUSIC — RESOLVE BLOCKER PROPERLY .11

.Current installation is advanced

:Already pinned

MOSS-Music source
MOSS-Music Instruct snapshot
MOSS-Music Thinking snapshot
SGLang moss-audio revision
Python 3.12
CUDA 12.8
Torch 2.9.1
FFmpeg 7

:Current blocker

MOSS-Music → TorchCodec 0.9 line
moss-audio SGLang → TorchCodec 0.8 line

HARD RULE

:Do NOT

pip install --force
ignore dependency

pick whichever installs last
monkey patch TorchCodec blindly

Required resolution

:Find a compatible reviewed pair

MOSS-Music source revision

+

SGLang moss-audio revision

+

TorchCodec version

:Acceptable resolution must satisfy

pip check = PASS

native TorchCodec load = PASS

tiny WAV decode = PASS

model load = PASS

:Then

Instruct smoke

:Input

.real 20–30 sec music

:Ask structured questions

tempo

key

structure

instrumentation

style

.Require meaningful response

Thinking smoke

.Same audio with reasoning-oriented prompt

.Require meaningful output

Persist proof separately

MOSS_MUSIC_INSTRUCT

MOSS_MUSIC_THINKING

Deploy

:Canonical

MOSS_MUSIC_API_URL

Final

MOSS_MUSIC_INSTRUCT = READY

MOSS_MUSIC_THINKING = READY

:If no compatible upstream resolution exists

BLOCKED_UPSTREAM

.Do NOT fake readiness

LADA-BAND — COMPLETE RESEARCH INSTALL .12

.Current worker exists

:Pinned

TME-LaDA-Band source

LaDA-Band HF snapshot

L40S

gated weights

:License

NONCOMMERCIAL_RESEARCH_ONLY

Required

.Confirm gated HF access .1

:Use Secret .2

LADA_BAND_HF_TOKEN

:Download .3

/checkpoints

/pretrained

:Create complete inventory .4

file

bytes

sha256

source revision

.Run a real vocal input .5

.Generate accompaniment .6
:Validate .7
WAV decodes -
not silent -
duration plausible -
not identical/source-copy -
.Persist smoke proof .8
.Deploy research endpoint .9
:Configure .10

LADA_BAND_API_URL

.Keep production router prohibition .11

:Final

LADA_BAND = RESEARCH_READY

.Never "READY" for commercial production

HAFM — COMPLETE .13

.Current worker and exact model/source revisions exist

Required

.Bootstrap source .1
.Download model snapshot .2
.Hash complete model tree .3
:Verify config .4
configs/ar.yaml
.Build runtime .5
.Run real isolated vocal .6
:Execute .7
infer_simple.py
:Validate generated instrumental .8
audible -
finite -
non-zero duration -
not source copy -
.Persist smoke .9
.Deploy L40S endpoint .10
:Configure .11

HAFM_API_URL

.Connect provider .12

:Final

HAFM = READY

.assuming Apache-2.0 source/weight licensing remains verified

ANYACCOMP — FIX THE NULL REVISIONS FIRST .14

:Current manifest problem

source.revision = null

weights.revision = null

.This must be fixed BEFORE downloading production assets

Required

Source

.Resolve full immutable Amphion/AnyAccomp source revision

:Record

repository

char commit SHA-40

subdirectory

HF model

:Resolve exact

amphion/anyaccomp

.snapshot SHA

:Download all

VQ

Flow Matching

Vocoder

configs

supporting assets

Runtime

:Keep isolated

Python 3.9

Torch 2.3.1

Torchaudio 2.3.1

Torchvision 0.18.1

Transformers 4.47.1

Accelerate 1.6.0

Smoke

:Input

real vocal

:Output

accompaniment WAV

:Require

non-silent

not input copy

valid duration

Deployment

:Configure

ANYACCOMP_API_URL

Licensing

.Do NOT assume commercial authorization

:If commercial rights are not verified

ANYACCOMP = RESEARCH_READY

:or

BLOCKED_LICENSE

.according to exact upstream terms

MIDI-SAG + MUSECONTROLLITE — COMPLETE THE FULL ASSET CHAIN .15

.Current worker already exists and source is pinned

.The current weak point is asset provenance

.All submodels currently need exact identities

:Required components

RMVPE
GAME-1.0-medium
MuseControlLite
Vocal Beat Tracking
AccoMontage2
CSLL2M
SoulX-Singer
SoulX-Singer-Preprocess

:For EACH asset replace generic descriptions with

repository/source URL
exact revision if available
filename
bytes
SHA256
license
download method

:Google Drive artifact

.SHA256 becomes identity authority

.Do NOT run upstream install.sh blindly

:Recreate every download in

bootstrap_assets.py

Smoke tests

Smoke A — individual MuseControlLite

:Provide

melody control
chroma
rhythm

dynamics

.Generate audible audio

Smoke B — detected mode

:Run

vocal

beat →

MIDI →

harmony →

MuseControlLite →

accompaniment →

:Require

valid intermediate MIDI

valid harmony

audible final WAV

Deploy

:Configure

MIDI_SAG_API_URL

:Also expose capability

MUSE_CONTROL_LITE

.even if same worker

:Final

MIDI_SAG = READY or RESEARCH_READY according to subasset licenses

MUSE_CONTROL_LITE = same license classification

MUSICGEN — FINISH EXACT MODEL PINNING .16

.Worker is already built

:Current blocker

resolved_revision = null

.for both MusicGen models

:Resolve immutable HF revisions for

facebook/musicgen-large
facebook/musicgen-melody-large

.Do NOT use short requested hashes only

.Store full resolved SHA

Provision

.Download all required model files

Environment

:Keep

Python 3.9
Torch 2.1.0+cu118
CUDA 11.8
FFmpeg 4.4

:The runtime must compile native

PyAV
PESQ

.before downloading large snapshots

Smoke

MusicGen Large

.Text prompt → WAV

MusicGen Melody Large

.melody/reference → generated WAV

:Require

non-silent
finite
valid sample rate
different from input

Deploy

:Configure

MUSICGEN_API_URL

License

.Current MusicGen weights are non-commercial

:Final

MUSICGEN_LARGE = RESEARCH_READY
MUSICGEN_MELODY_LARGE = RESEARCH_READY

JASCO — KEEP BLOCKED UNTIL IMMUTABLE WEIGHTS ARE REVIEWED .17

:Current

BLOCKED_NO_WEIGHTS

:Do NOT resolve

facebook/jasco main

.automatically

:Required before installation

exact checkpoint repository
immutable revision
SHA256
license

:Only then

.install inside existing AudioCraft worker

:If not resolvable

JASCO_CHORDS_DRUMS_MELODY = BLOCKED_NO_WEIGHTS

.This counts as truthful completion

DIFFRHYTHM 2 — EXECUTE THE REAL SMOKE .18

.Worker is already well built

:Pinned

DiffRhythm2 source
ASLP-lab/DiffRhythm2
MuQ-MuLan dependency
Apache-2.0

:A real smoke function already exists using

שמוליק סוכות - הילד הזה

.second fixture-30

Required now

:RUN

()smoke_real_audio

.Do not just keep function code

:Verify persisted proof

realInference=true
nonSilent=true
notSourceCopy=true
lyricsConditioned=true
rhythmConditioned=true

Then deploy real endpoint

:Configure

DIFFRHYTHM2_API_URL

.Health must validate proof

:Final

DIFFRHYTHM_2 = READY

STABLE AUDIO 3 — FINISH DEPLOYMENT LAYER .19

:Current worker has

app
inference
bootstrap
manifest
pyproject
tests

.but deployment stack is incomplete

Add

Dockerfile
modal_config.py
modal_app.py
modal_provision.py
smoke.py
license_manifest.json

:Provision both

STABLE_AUDIO_3_SMALL_MUSIC
STABLE_AUDIO_3_MEDIUM

.Keep exact existing HF revisions

Runtime

.Pin exact environment

:Do not leave broad unbounded

...=<transformers

.in production lockfile

.Resolve exact versions during lock

Smoke tests

Small

.Text-to-audio

Medium

.Text-to-audio

Audio-to-audio

.Existing WAV → transformed WAV

Inpainting

.Use known input

.Regenerate one exact interval

:Validate

outside region preserved sufficiently

inside region changed

output valid

Continuation

.Provide audio prefix and generate continuation

Deploy

:Canonical

STABLE_AUDIO_3_API_URL

:Final

STABLE_AUDIO_3_SMALL_MUSIC = READY

STABLE_AUDIO_3_MEDIUM = READY

.subject to Stability license acceptance

NOW BEGIN PROVIDERS THAT HAVE NOT BEEN STARTED .20

.Only after sections 6–19 are closed

SAM AUDIO .21

:Create

/services/sam-audio-worker

:Required files

Dockerfile
requirements.lock
model_manifest.json
license_manifest.json
bootstrap_assets.py
app.py
inference.py
smoke.py
modal_config.py
modal_app.py
modal_provision.py
/tests

Source

:Pin exact

facebookresearch/sam-audio

.revision

Runtime

.Use isolated Python >=3.11

.Do not share dependency tree with other workers

Models

:Provision reviewed checkpoints

:Start with

sam-audio-base
sam-audio-large

.Small optional

Gated assets

.Use Secret token only

Smoke tests

.Use mixed music

:Prompts

"vocals"

"acoustic guitar"

"drums"

:Each output must

decode

be audible

differ from residual

Capabilities

TEXT_PROMPT_SEPARATION

TEMPORAL_PROMPT_SEPARATION

OPEN_VOCAB_SEPARATION

:Deploy

SAM_AUDIO_API_URL

:Final

:according to exact SAM license

READY

.or license-limited status if required

MAGENTA REALTIME 2 .22

:Create

/services/magenta-rt2-worker

:Install both

MRT2_SMALL

MRT2_BASE

:Assets

MusicCoCa

SpectroStream

mrt2_small
mrt2_base

Runtime

:Reference

Python 3.12
JAX CUDA runtime

.Pin exact package versions

Smoke A

.Text-conditioned music

Smoke B

.MIDI-conditioned generation

Smoke C

.Audio context continuation

Verify

.MIDI must cause output difference

.Do not claim deterministic rendering

:Provider metadata

GENERATIVE_MIDI_CONDITIONED

Deploy

MAGENTA_RT2_API_URL

:License

code Apache-2.0
weights CC-BY-4.0

:Final

MAGENTA_RT2_SMALL = READY
MAGENTA_RT2_BASE = READY

MUSIC2MUSIC + REMI-z .23

:Create

/services/music2music-worker

:Install

REMI-z

.Pin exact source

:Models

m2m_arranger
m2m_pianist_dur
m2m_drummer
m2m_pt

.Resolve full HF revision for every model

Smoke

:Input

known MIDI

:Output

valid multitrack MIDI

:Require

non-empty tracks -
finite note events -
changed arrangement -
valid instrument information -

License

:If public model license still unclear

:Final

MUSIC2MUSIC = RESEARCH_READY

.only if research use is legally supported

:Otherwise

BLOCKED_LICENSE

.Do NOT mark production ready

METEOR .24

:Create

/services/meteor-worker

.Pin exact source revision

:Environment

Python 3.9.2

.Install exact requirements

:Use

no-deps--

.where upstream requires

:Provision

official checkpoint

tokenizer

corpus/preprocessing assets

REMI/MuseMorphose support

Smoke

:Known MIDI

MIDI

events →

model →

output MIDI →

.Validate actual note changes

Deploy

METEOR_API_URL

:Final

:license-dependent

READY

:or

RESEARCH_READY

SYMPHONYGEN .25

:Create

/services/symphonygen-worker

:Environment

Python 3.11

torch==2.8.0

torchaudio==2.8.0

transformers==5.1.0

:Provision

stage_one_pretrained.pt

stage_two_pretrained.pt

grpo_clamp_epoch_10.pt

grpo_clamp+track_epoch_6.pt

.Hash all

.Optional assets needed only for evaluation/training should be marked optional

Smoke

.MIDI/conditioning → multi-track orchestral MIDI

:Require

multiple tracks =<

real notes =<

valid SMF

:Deploy

SYMPHONYGEN_API_URL

MuQ SHARED EMBEDDING SERVICE .26

:Create

/services/music-embedding-worker

:Providers

MUQ

MUQ_MULAN

.Do not duplicate assets unnecessarily

Capabilities

audio embedding

text embedding

music-text similarity

music-music similarity

section embeddings

Smoke

:Same song vs unrelated song

:Require

similar-song score > unrelated score

.for a controlled fixture

:License

.Record CC-BY-NC where applicable

:Final

:likely

RESEARCH_READY

MuQ-Eval .27

:Create

/services/muq-eval-worker

:Model

MuQ-Eval-A1

.Use exact checkpoint revision

Smoke

:Generate evaluation score for

clean fixture
degraded fixture

.The degraded fixture should normally score lower

.Do not hardcode score values

Output

:Canonical

```
    }  
    ,qualityScore": 0.0"  
    ,"" : "model"  
    {} : "provenance"  
    {
```

:Final

MUQ_EVAL = READY

.subject to verified model license

SONG AESTHETICS EVALUATION .28

:Create

/services/song-aesthetics-worker

:Input

mixture

vocal stem

accompaniment stem

:Output

Overall

Singing

Melody

Arrangement

AudioQuality

.Pin exact source/checkpoint

Smoke

.Known clean and degraded music

.Verify valid 0–100 outputs

:Final

.license dependent

SONGEVAL .29

:Create

/services/song-eval-worker

:License

CC BY-NC-SA 4.0

:Therefore

RESEARCH_READY

.never production commercial

.Provision exact model assets

.Run real song inference

CLaMP 3 .30

:Create optional

/services/clamp3-worker

:Capabilities

text ↔ audio

audio ↔ MIDI

text ↔ MIDI

score ↔ audio

.Provision exact model

Smoke

.Known matching pair should score higher than mismatched pair

Final

:Mark

RESEARCH_READY

.unless exact weights license says otherwise

sfizz + VSCO2 CE .31

.Current API infrastructure exists

.Now provision actual native assets

Install sfizz

:Pin

sfizz 1.2.3

.Build actual native MIDI render host

Provision VSCO2 CE

:Path outside Git

/var/lib/music-ai/assets/sfz/vsco2-ce/

:Calculate deterministic tree hash

SHA256

file count

bytes

version

Smoke

.Render canonical TrackModel

:Must respond to

note pitch

velocity

CC

articulation

automation

.Render at least 3 different TrackModels

.Output hashes must differ

:Final

SFIZZ_VSCO2_CE = READY

VST3 NATIVE HOST .32

.The application infrastructure already supports VST3

.Now finish actual native host

:Host must support

MIDI notes

velocity

CC

pitch bend

automation

plugin state

offline render

Host security

:Exact host binary must be approved by

MUSIC_AI_APPROVED_NATIVE_HOSTS

:with

identity

SHA256

Licensed VST3 instrument

.Do NOT download unlicensed plugin

:If user has not supplied licensed instrument

:Final

VST3_HOST = READY

VST3_INSTRUMENT = BLOCKED_MISSING_LICENSED_ASSET

.This counts as correct completion

LEVO2 RESEARCH .33

.Only after all higher-priority installs

.Create research-only worker

.Do not include in production routing

:License

research / academic only

:Final

LEVO2 = RESEARCH_READY

.if private research terms are satisfied

:Otherwise

BLOCKED_LICENSE

WATCHLIST — DO NOT INSTALL FAKE PROVIDERS .34

:Keep

S2Accompanist

BeatEdit

Live Music Diffusion Models

Diff-A-Riff

Break-the-Beat

:in

research-watchlist.json

:Schema

}

, "" : "provider"

, "" : "paper"

, "" : "repository"

, weightsAvailable": false"

, licenseKnown": false"

, "status": "BLOCKED_NO_WEIGHTS"

"" : "lastReviewed"

{

.Do NOT create runtime or Docker worker without real public installable checkpoint

REQUIRED PROVIDER FILE CONTRACT .35

:Every provider must contain where relevant

Dockerfile

requirements.lock / uv.lock

model_manifest.json

license_manifest.json

bootstrap_assets.py

app.py

inference.py

smoke.py

README.md

modal_config.py

```
modal_app.py
modal_provision.py
/tests
```

.Missing required file means install incomplete

MODEL MANIFEST STANDARD .36

:Every "model_manifest.json" must contain

```
    }
    , "" : "provider"
      } : "code"
    , "" : "repository"
      "" : "revision"
    , {
      } : "runtime"
    , "" : "python"
    , "" : "cuda"
    , "" : "torch"
    , "" : "torchaudio"
    , "" : "torchvision"
    "" : "transformers"
    , {
      , [] : "models"
      , [] : "assets"
      , "" : "assetRoot"
      , "" : "smokeProof"
      {} : "license"
    }
  }
```

:Every asset

```
    }
    , "" : "repository"
    , "" : "revision"
      , "" : "path"
      , bytes": 0"
      "" : "sha256"
    }
```

IMMUTABILITY RULE .37

:Forbidden

revision=main
revision=master
latest
null

.in final installation matrix for installable components

:If upstream does not expose immutable revision

.artifact SHA256 must be recorded, and the limitation documented explicitly

HUGGING FACE RULE .38

:No production runtime

("from_pretrained("repo

.without local/offline provisioned snapshot

.All HF downloads happen during provisioning

:Production

HF_HUB_OFFLINE=1
TRANSFORMERS_OFFLINE=1

.where compatible

NO REQUEST-TIME DOWNLOADS .39

:Runtime must never download

checkpoint
tokenizer
codec
VAE
vocoder
model config
Git repo
Google Drive artifact
HF snapshot

:If missing

.return unavailable

.Do not auto-download

VOLUME STRUCTURE .40

:Recommended

/music-ai-models-v2

/ace-step

/bs-roformer

/mt3

/mr-mt3

/your-mt3

/lada-band

/hafm

/anyaccomp

/midi-sag

/musicgen

/stable-audio3

/diffrrhythm2

/moss

/magenta-rt2

/sam-audio

/music2music

/meteor

/symphonygen

/levo2

/music-analysis-models-v2

/beat-this

/songformer

/sheetsage

/muq

/muq-eval

/song-aesthetics

/songeval

/clamp3

/music-assets-v2

/sfz

/vst3

SECRET INVENTORY .41

:Create/document only required Secrets

MUSIC_AI_WORKER_TOKEN

HF_TOKEN

LADA_BAND_HF_TOKEN

STABILITY_AI_HF_TOKEN

SAM_AUDIO_HF_TOKEN

SHEETSAGE_API_TOKEN

SHEETSAGE_LICENSE_SECRET

provider-specific API tokens

promotion signing private key

.Never put them in Git or ".replit" plaintext

.Public keys may be stored when appropriate

API ENVIRONMENT CONTRACT .42

:Final API should have canonical endpoint env keys for installed services

ACE_STEP_API_URL

BS_ROFORMER_API_URL

MT3_API_URL

MR_MT3_API_URL

YOUR_MT3_API_URL

ALL_IN_ONE_API_URL

SONGFORMER_API_URL

BEAT_THIS_API_URL

SHEETSAGE_API_URL

MOSS_MUSIC_API_URL

LADA_BAND_API_URL

HAFM_API_URL

ANYACCOMP_API_URL

MIDI_SAG_API_URL

MUSICGEN_API_URL
STABLE_AUDIO_3_API_URL
DIFFRHYTHM2_API_URL

SAM_AUDIO_API_URL
MAGENTA_RT2_API_URL

MUSIC2MUSIC_API_URL
METEOR_API_URL
SYMPHONYGEN_API_URL

MUQ_API_URL
MUQ_EVAL_API_URL
SONG_AESTHETICS_API_URL
SONG_EVAL_API_URL
CLAMP3_API_URL

.Use one canonical spelling

.Legacy aliases only temporarily

INSTALLATION EXECUTION ORDER .43

PHASE 1 — CLOSE ALMOST-DONE

:Execute exactly

SheetSage .1
Beat This .2
MR-MT3 .3
YourMT3 .4
SongFormer .5

.Do not start Phase 2 until these are final-status

PHASE 2 — CLOSE NEW HIGH-VALUE STACK

MOSS-Music .6
LaDA-Band .7
HAFM .8
AnyAccomp .9
MIDI-SAG + MuseControlLite .10

PHASE 3 — GENERATION STACK

MusicGen .11
JASCO status resolution .12
DiffRhythm2 .13
Stable Audio 3 Small .14
Stable Audio 3 Medium .15

PHASE 4 — NEW CAPABILITIES

SAM Audio .16
Magenta RT2 Small .17
Magenta RT2 Base .18

PHASE 5 — SYMBOLIC STACK

REMI-z .19
Music2Music .20
METEOR .21
SymphonyGen .22

PHASE 6 — EVALUATION / EMBEDDINGS

MuQ .23
MuQ-MuLan .24
MuQ-Eval .25
Song Aesthetics .26
SongEval .27
CLaMP3 .28

PHASE 7 — RENDERERS

sfizz .29
VSCO2 CE .30
VST3 native host .31
licensed VST3 asset status .32

PHASE 8 — OPTIONAL RESEARCH

LeVo2 research .33

AFTER EACH PROVIDER .44

:Immediately update

installation-matrix-v2.json

.Do not wait until the end

:Every READY status must link internally to

model manifest

asset manifest

smoke proof

endpoint

promotion record if applicable

FINAL AUDIT SCRIPT .45

:Create

scripts/audit-installation-stack.py

:It should read

installation-matrix-v2.json

.and fail if any provider has contradictions

:Examples

installed=true but assetsDownloaded=false

READY but smokePassed=false

READY but modelRevision=null

READY GPU provider but promotionSigned=false

READY but endpointConfigured=false

commercial READY with licenseStatus=RESEARCH_ONLY

:Output

PASS

.only if entire stack is truthful

GENERATE FINAL REPORT .46

:Create

docs/installation-completion-report.md

:Include table

Provider
Category
Version
Runtime
Model revision
Checkpoint SHA
License
Endpoint
Smoke result
Promotion
Final status
Blocker

.No prose-only summary

FINAL STACK EXPECTATION .47

:At end, the matrix should contain at least

- ACE_STEP
- BS_ROFORMER
- DEMUCS
- ALL_IN_ONE
- BEAT_THIS
- SONGFORMER
- SHEETSAGE
- MADMOM
- ESSENTIA
- CHROMA
- TORCHCREPE
- BASIC_PITCH
- PYLOUDNORM

MT3
MR_MT3
YOUR_MT3

MOSS_MUSIC_INSTRUCT
MOSS_MUSIC_THINKING

LADA_BAND
HAFM
ANYACCOMP
MIDI_SAG
MUSE_CONTROL_LITE

MUSICGEN_LARGE
MUSICGEN_MELODY_LARGE
JASCO_CHORDS_DRUMS_MELODY
STABLE_AUDIO_3_SMALL_MUSIC
STABLE_AUDIO_3_MEDIUM
DIFFRHATHM_2
LEVO2_RESEARCH

SAM_AUDIO

MAGENTA_RT2_SMALL
MAGENTA_RT2_BASE

REMI_Z
MUSIC2MUSIC
METEOR
SYMPHONYGEN

MUQ
MUQ_MULAN
MUQ_EVAL
SONG_AESTHETICS
SONG_EVAL
CLAMP3

PEDALBOARD
SFIZZ_VSCO2_CE
VST3_HOST
VST3_INSTRUMENT

SUCCESS CRITERION .48

:The installation phase is complete when

.There are zero providers in an undefined state

:Every selected provider is

READY
RESEARCH_READY
BLOCKED_LICENSE
BLOCKED_NO_WEIGHTS
BLOCKED_UPSTREAM
BLOCKED_MISSING_LICENSED_ASSET

:Every installable provider has

exact source
exact model
exact runtime
downloaded assets
hashes
durable storage
real smoke
endpoint
API configuration
license record

:And

python scripts/audit-installation-stack.py

:returns

PASS

STOP CONDITION .49

:When the installation audit passes

.STOP

.Do not start Golden Song automatically

.Do not start musical optimization

.Do not rewrite Arrangement Director

.Do not generate candidates

:The next task will explicitly be

REAL MUSICAL CERTIFICATION PHASE

:That phase will cover

provider-by-provider real inference

Golden Song

Song Model validation

generator shootout

candidate ranking

human listening

failure taxonomy

Arrangement Director V2

Phrase Planner

CompositionEngine V2

repaint / section repair

professional production quality

.But NONE of that belongs to this task

FINAL INSTRUCTION TO AGENT

.Work through the installation matrix systematically

.Do not optimize for number of commits

.Do not optimize for number of created workers

:Optimize only for one result

Every selected component has a truthful, reproducible, immutable and auditable installation»

«.state

.A Dockerfile is not an installation

.A manifest is not an installation

.A downloaded checkpoint is not an installation

.An HTTP 200 is not an installation

:A provider is complete only when the full chain exists

PIN
BUILD →
DOWNLOAD →
HASH →
PROVISION →
REAL INFERENCE →
VERIFY →
ATTEST →
DEPLOY →
CONNECT →
READY →

.Only then mark it complete